Segundo Examen Parcial

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1) datos: (3, 2, 3, 0, 3, 1, 3, 2, 3, 1)

$n = 10$ observaciones

x	centro
(0, 1)	
(1, 2)	
(2, 2)	
(3, 5)	

X	0	1	2	3
P	$3\theta/3$	$6\theta/3$	$\frac{1-3\theta}{3}$	$\frac{2(1-3\theta)}{3}$

$$L(\theta) = f(x,\theta)\, f(x_1,\theta) \ldots$$

$$L(\theta) = \left(\frac{3\theta}{3}\right)\left(\frac{6\theta}{3}\right)^2\left(\frac{1-3\theta}{3}\right)^2\left(\frac{2(1-3\theta)}{3}\right)^5$$

$$L(\theta) = \theta \cdot (2\theta)^2\left(\frac{1-3\theta}{3}\right)^2\left(\frac{2(1-3\theta)}{3}\right)^5$$

Constante

$$L(\theta) = \theta \cdot 4\theta^2 \cdot \left(\frac{1}{9}\right)(1-3\theta)^2 \cdot \left(\frac{32}{243}\right)(1-3\theta)^5$$

$x = \frac{128}{2187}$

$$L(\theta) = \frac{128}{2187}\theta^3(1-3\theta)^7$$

$$L(\theta) = x\theta^3(1-3\theta)^7$$

$$\frac{d}{d\theta}\, l(\theta) = \ln x + 3\ln(\theta) + 7\ln(1-3\theta)$$

$$\frac{d}{d\theta}\, l(\theta) = 0 + \frac{3}{\theta} + 7 \cdot (-3)/(1-3\theta)$$

$$\frac{d}{d\theta}\, l(\theta) = \frac{3}{\theta} - \frac{21}{(1-3\theta)} = 0$$

$$\frac{3}{\theta} - \frac{21}{(1-3\theta)} = 0$$

$$\frac{3}{\theta} = \frac{21}{(1-3\theta)}$$

$$3(1-3\theta) = 21\theta$$

$$3 - 9\theta = 21\theta$$

$$3 = 21\theta + 9\theta$$

$$3 = 30\theta$$

$$\theta = \frac{3}{30} = 0.1$$

3)

Priori: Distribución beta $\qquad \beta(1,3)$

$$f(p \mid \alpha, \beta) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha,\beta)}, \quad p \in [0,1]$$

$$f(p) = \frac{p^{1-1}(1-p)^{3-1}}{B(1,3)}$$

$$\frac{(1-p)^2}{\beta(1,3)} \propto (1-p)^2$$

$k=3$ morosos

$n=8$ clientes

$$L(p \mid k,n) = \binom{n}{k} p^k (1-p)^{n-k} \longrightarrow \text{Conjunta}$$

$$\binom{8}{3} p^3 (1-p)^5$$

$$L(p) \propto p^3 (1-p)^5$$

Posterior: Inferencia Bayesiana

$$f(p \mid \text{datos}) \propto L(p \mid \text{datos}) \cdot f(p)$$

$$f(p \mid \text{datos}) \propto p^3(1-p)^5 \cdot (1-p)^2 = p^3(1-p)^7$$

$\alpha - 1 = 3$

$\alpha = 4$ $\qquad p \mid \text{datos} \approx \beta(4,8)$

$\beta - 1 = 7$

$\beta = 8$

Media

$$E_p = \frac{\alpha}{\alpha+\beta}$$

$$\frac{4}{4+8}$$

$$\frac{1}{3} \quad 0.333$$

Varianza $= \dfrac{\alpha \cdot \beta}{(\alpha+\beta)^2 \cdot (\alpha+\beta+1)}$

$$\frac{4 \cdot 8}{(4+8)^2 (4+8+1)}$$

$$\frac{32}{144 \cdot 13}$$

$$\frac{2}{117} = 0.0171$$